Number theory – 4

p-adic valuations, factorials and binomial coefficients

Homework

14 сентября 2026 года

Theoretical minimum. The valuation $v_p(n)$ is the exponent of p in n; it is additive on products and satisfies $v_p(x+y) \geqslant \min(v_p(x), v_p(y))$ with equality when the valuations differ. Legendre's formula gives $v_p(n!) = \sum_{i \geqslant 1} \lfloor n/p^i \rfloor = (n - s_p(n))/(p-1)$ with s_p the digit sum in base p. Kummer's theorem says $v_p\binom{n}{k}$ is the number of carries when k and $n-k$ are added in base p; hence all $\binom{n}{k}$ are odd exactly for $n = 2^m - 1$, and $\binom{2n}{n}$ has no prime factor between $2n/3$ and n. Lifting the exponent: for odd p with $p \mid a - b$ and $p \nmid a$, $v_p(a^n - b^n) = v_p(a-b) + v_p(n)$; for $p = 2$ and even n there is the extra term $v_2(a+b) - 1$. Lucas' theorem is in Combinatorics – 3, Stirling's formula in Limits – 6, Chebyshev's bounds in Series and infinite products – 6.

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1. Prove Erdős's estimate $\prod_{p \leqslant n} p < 4^n$ by induction using $\binom{2m+1}{m}$, and combine it with Kummer's theorem to prove Bertrand's postulate: for every $n \geqslant 1$ there is a prime between n and $2n$.

2. Prove that $\binom{2n}{n} \mid \operatorname{lcm}(1, 2, \ldots, 2n)$, that $(n+1)\binom{n}{k} \mid \operatorname{lcm}(1, \ldots, n+1)$ for $0 \leqslant k \leqslant n$, and that $\operatorname{lcm}(1, \ldots, 2m) \mid \binom{2m}{m} \operatorname{lcm}(1, \ldots, m)$; deduce the Chebyshev-type bounds $2^n \leqslant \operatorname{lcm}(1, 2, \ldots, n+1) \leqslant 4^n$ for $n \geqslant 1$, and a lower bound for the number of primes up to $2n$.

3. Determine all positive integers n such that $v_2\big(\binom{2n}{n}\big) = 1$, deduce that $\binom{2n}{n}$ can be squarefree only when n is a power of 2, and find all $n \leqslant 100$ for which $\binom{2n}{n}$ is squarefree.

4. Let p be an odd prime. Prove that $v_p\big(\sum_{k=1}^{p-1} \frac{1}{k}\big) \geqslant 2$ and $v_p\big(\sum_{k=1}^{p-1} \frac{1}{k^2}\big) \geqslant 1$ for $p \geqslant 5$, and compute both valuations for $p = 3$.

5. Let $a_k = \lfloor n^k/k \rfloor$ for a fixed integer $n > 1$. Prove that infinitely many terms of the sequence $(a_k)_{k \geqslant 1}$ are odd. (IMO shortlist 2014, N4)

6. For each positive integer k let $t(k)$ be the largest odd divisor of k. Determine all positive integers a for which there exists a positive integer n such that all the differences $t(n+a)-t(n)$, $t(n+a+1)-t(n+1)$, $\ldots$, $t(n+2a-1)-t(n+a-1)$ are divisible by 4. (IMO shortlist 2011, N4)

7. For $d > 0$ let $f(d)$ be the smallest positive integer with exactly d divisors. Prove that $f(2^{k+1})$ is divisible by $f(2^k)$ for every k.
(IMO shortlist 2011, N1)

8. Prove that $\binom{2n}{n}$ is never a perfect power for $n > 1$, using Bertrand's postulate and Kummer's theorem.

9. Prove that for every n the number $\operatorname{lcm}\left(\binom{n}{0}, \binom{n}{1}, \ldots, \binom{n}{n}\right)$ equals $\operatorname{lcm}(1, 2, \ldots, n+1)/(n+1)$, and deduce the identity $\sum_k \frac{(-1)^k}{\binom{n}{k}}$ in closed form.

10. Let p be a prime and let n be written in base p. Prove that the exponent of p in the multinomial coefficient $\frac{n!}{k_1! \cdots k_m!}$ equals the number of carries in the base-p addition of $k_1, \ldots, k_m$, and use it to determine when a multinomial coefficient is coprime to p.